Algoma UNIVERSITY

School of Computer Science & Technology - Algoma

Linear Algebra

Math 1057

Instructor

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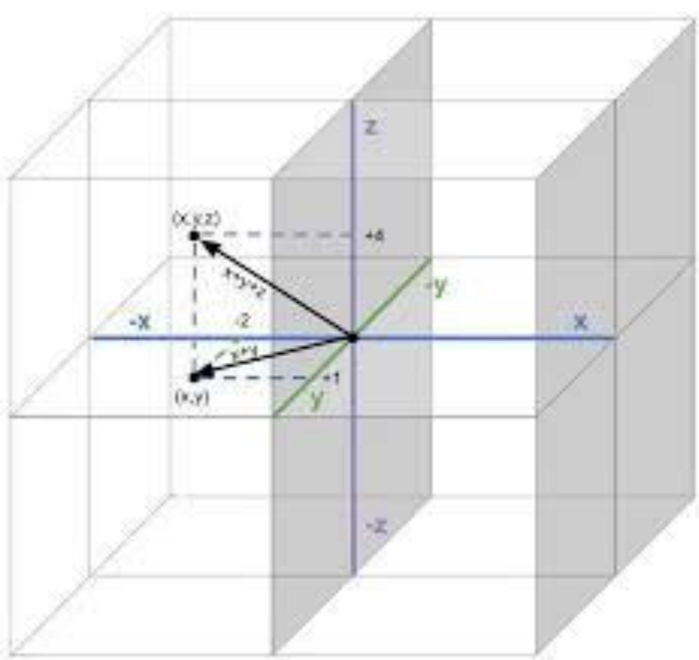

16 Linear Independence, Bases, and Dimension

Lecture 16

Linear Independence, Bases, and Dimension

16.1 Linear Independence

Roughly speaking, the concept of linear independence evolves around the idea of working with "efficient" spanning sets for a subspace. For instance, the set of directions

$$\{\text{EAST}, \text{NORTH}, \text{NORTH-EAST}\}$$

are redundant since a total displacement in the NORTH-EAST direction can be obtained by combining individual NORTH and EAST displacements. With these vague statements out of the way, we introduce the formal definition of what it means for a set of vectors to be "efficient".

Definition 16.1: Let V be a vector space and let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ be a set of vectors in V. Then $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ is **linearly independent** if the only scalars $c_1, c_2, \ldots, c_p$ that satisfy the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$$

are the trivial scalars $c_1 = c_2 = \cdots = c_p = 0$. If the set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is not linearly independent then we say that it is **linearly dependent**.

We now describe the redundancy in a set of linear dependent vectors. If $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ are linearly dependent, it follows that there are scalars $c_1, c_2, \ldots, c_p$, **at least one of which is nonzero**, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p = \mathbf{0}. \qquad (\star)$$

For example, suppose that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ are linearly dependent. Then there are scalars c_1, c_2, c_3, c_4, not all of them zero, such that equation $(\star)$ holds. Suppose, for the sake of argument, that $c_3 \neq 0$. Then,

$$\mathbf{v}_3 = -\frac{c_1}{c_3}\mathbf{v}_1 - \frac{c_2}{c_3}\mathbf{v}_2 - \frac{c_4}{c_3}\mathbf{v}_4.$$

Therefore, when a set of vectors is linearly dependent, it is possible to write one of the vectors as a linear combination of the others. It is in this sense that a set of linearly dependent vectors are redundant. In fact, if a set of vectors are linearly dependent we can say even more as the following theorem states.

Theorem 16.2: A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$, with $\mathbf{v}_1 \neq \mathbf{0}$, is linearly dependent if and only if some $\mathbf{v}_j$ is a linear combination of the preceding vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$.

Example 16.3. Show that the following set of 2×2 matrices is linearly dependent:

$$\left\{ \mathbf{A}_1 = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix},\ \mathbf{A}_2 = \begin{bmatrix} -1 & 3 \\ 1 & 0 \end{bmatrix},\ \mathbf{A}_3 = \begin{bmatrix} 5 & 0 \\ -2 & -3 \end{bmatrix} \right\}.$$

Solution. It is clear that $\mathbf{A}_1$ and $\mathbf{A}_2$ are linearly independent, i.e., $\mathbf{A}_1$ cannot be written as a scalar multiple of $\mathbf{A}_2$, and vice-versa. Since the $(2,1)$ entry of $\mathbf{A}_1$ is zero, the only way to get the -2 in the $(2,1)$ entry of $\mathbf{A}_3$ is to multiply $\mathbf{A}_2$ by -2. Similary, since the $(2,2)$ entry of $\mathbf{A}_2$ is zero, the only way to get the -3 in the $(2,2)$ entry of $\mathbf{A}_3$ is to multiply $\mathbf{A}_1$ by 3. Hence, we suspect that $3\mathbf{A}_1 - 2\mathbf{A}_2 = \mathbf{A}_3$. Verify:

$$3\mathbf{A}_1 - 2\mathbf{A}_2 = \begin{bmatrix} 3 & 6 \\ 0 & -3 \end{bmatrix} - \begin{bmatrix} -2 & 6 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ -2 & -3 \end{bmatrix} = \mathbf{A}_3$$

Therefore, $3\mathbf{A}_1 - 2\mathbf{A}_2 - \mathbf{A}_3 = \mathbf{0}$ and thus we have found scalars c_1, c_2, c_3 not all zero such that $c_1\mathbf{A}_1 + c_2\mathbf{A}_2 + c_3\mathbf{A}_3 = \mathbf{0}$. $\square$

16.2 Bases

We now introduce the important concept of a basis. Given a set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_{p-1}, \mathbf{v}_p\}$ in V, we showed that $\mathsf{W} = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ is a subspace of V. If say $\mathbf{v}_p$ is linearly dependent on $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{p-1}$ then we can remove $\mathbf{v}_p$ and the smaller set $\{\mathbf{v}_1, \ldots, \mathbf{v}_{p-1}\}$ still spans all of W:

$$\mathsf{W} = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{p-1}, \mathbf{v}_p\} = \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_{p-1}\}.$$

Intuitively, $\mathbf{v}_p$ does not provide an independent "direction" in generating W. If some other vector $\mathbf{v}_j$ is linearly dependent on $\mathbf{v}_1, \ldots, \mathbf{v}_{p-1}$ then we can remove $\mathbf{v}_j$ and the resulting smaller set of vectors still spans W. We can continue removing vectors until we obtain a minimal set of vectors that are linearly independent and still span W. The following remarks motivate the following important definition.

Definition 16.4: Let W be a subspace of a vector space V. A set of vectors $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ in W is said to be a **basis** for W if

(a) the set $\mathcal{B}$ spans all of W, that is, $\mathsf{W} = \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$, and

(b) the set $\mathcal{B}$ is linearly independent.

A basis is therefore a **minimal** spanning set for a subspace. Indeed, if $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is a basis for W and we remove say $\mathbf{v}_p$, then $\tilde{\mathcal{B}} = \{\mathbf{v}_1, \ldots, \mathbf{v}_{p-1}\}$ cannot be a basis for W. Why? If $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is a basis then it is linearly independent and therefore $\mathbf{v}_p$ **cannot** be written as a linear combination of the others. In other words, $\mathbf{v}_p \in \mathsf{W}$ is not in the span of $\tilde{\mathcal{B}} = \{\mathbf{v}_1, \ldots, \mathbf{v}_{p-1}\}$ and therefore $\tilde{\mathcal{B}}$ is not a basis for W because a basis must be a spanning set. If, on the other hand, we start with a basis $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ for W and we add a new vector $\mathbf{u}$ from W then $\tilde{\mathcal{B}} = \{\mathbf{v}_1, \ldots, \mathbf{v}_p, \mathbf{u}\}$ is not a basis for W. Why? We still have that $\operatorname{span} \tilde{\mathcal{B}} = \mathsf{W}$ but now $\tilde{\mathcal{B}}$ is not linearly independent. Indeed, because $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is a basis for W, the vector $\mathbf{u}$ can be written as a linear combination of $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$, and thus $\tilde{\mathcal{B}}$ is not linearly independent.

Example 16.5. Show that the standard unit vectors form a basis for $\mathsf{V} = \mathbb{R}^3$:

$$\mathbf{e}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix}0\\1\\0\end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

Solution. Any vector $\mathbf{x} \in \mathbb{R}^3$ can be written as a linear combination of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$:

$$\mathbf{x} = \begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = x_1 \begin{bmatrix}1\\0\\0\end{bmatrix} + x_2 \begin{bmatrix}0\\1\\0\end{bmatrix} + x_3 \begin{bmatrix}0\\0\\1\end{bmatrix} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3$$

Therefore, $\operatorname{span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \mathbb{R}^3$. The set $\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is linearly independent. Indeed, if there are scalars c_1, c_2, c_3 such that

$$c_1\mathbf{e}_1 + c_2\mathbf{e}_2 + c_3\mathbf{e}_3 = \mathbf{0}$$

then clearly they must all be zero, $c_1 = c_2 = c_3 = 0$. Therefore, by definition, $\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is a basis for $\mathbb{R}^3$. This basis is called the **standard basis** for $\mathbb{R}^3$. Analogous arguments hold for $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ in $\mathbb{R}^n$. □

Example 16.6. Is $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ a basis for $\mathbb{R}^3$?

$$\mathbf{v}_1 = \begin{bmatrix}2\\0\\-4\end{bmatrix}, \ \mathbf{v}_2 = \begin{bmatrix}-4\\-2\\8\end{bmatrix}, \ \mathbf{v}_3 = \begin{bmatrix}4\\-6\\-6\end{bmatrix}$$

Solution. Form the matrix $\mathbf{A} = [\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3]$ and row reduce:

$$\mathbf{A} \sim \begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$$

Therefore, the only solution to $\mathbf{Ax} = \mathbf{0}$ is the trivial solution. Therefore, $\mathcal{B}$ is linearly independent. Moreover, for any $\mathbf{b} \in \mathbb{R}^3$, the augmented matrix $\begin{bmatrix}\mathbf{A} & \mathbf{b}\end{bmatrix}$ is consistent. Therefore, the columns of $\mathbf{A}$ span all of $\mathbb{R}^3$:

$$\mathrm{Col}(\mathbf{A}) = \mathrm{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \mathbb{R}^3.$$

Therefore, $\mathcal{B}$ is a basis for $\mathbb{R}^3$. □

Example 16.7. In $\mathsf{V} = \mathbb{R}^4$, consider the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \\ 0 \\ -2 \end{bmatrix}, \ \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ -2 \\ 1 \end{bmatrix}, \ \mathbf{v}_3 = \begin{bmatrix} -1 \\ 4 \\ 2 \\ -3 \end{bmatrix}.$$

Let $\mathsf{W} = \mathrm{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. Is $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ a basis for W?

Solution. By definition, $\mathcal{B}$ is a spanning set for W, so we need only determine if $\mathcal{B}$ is linearly independent. Form the matrix, $\mathbf{A} = [\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3]$ and row reduce to obtain

$$\mathbf{A} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence, $\mathrm{rank}(\mathbf{A}) = 2$ and thus $\mathcal{B}$ is linearly dependent. Notice $\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{v}_3$. Therefore, $\mathcal{B}$ is not a basis of W. □

Example 16.8. Find a basis for the vector space of 2×2 matrices.

Example 16.9. Recall that a $n \times n$ is skew-symmetric $\mathbf{A}$ if $\mathbf{A}^T = -\mathbf{A}$. We proved that the set of $n \times n$ matrices is a subspace. Find a basis for the set of 3×3 skew-symmetric matrices.

16.3 Dimension of a Vector Space

The following theorem will lead to the definition of the dimension of a vector space.

Theorem 16.10: Let V be a vector space. Then all bases of V have the same number of vectors.

Proof: We will prove the theorem for the case that $\mathsf{V} = \mathbb{R}^n$. We already know that the standard unit vectors $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ is a basis of $\mathbb{R}^n$. Let $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ be nonzero vectors in $\mathbb{R}^n$ and suppose first that $p > n$. In Lecture 6, Theorem 6.7, we proved that any set of vectors in $\mathbb{R}^n$ containing more than n vectors is automatically linearly dependent. The reason is that the RREF of $\mathbf{A} = \begin{bmatrix}\mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_p\end{bmatrix}$ will contain at most $r = n$ leading ones, and therefore $d = p - n > 0$. Therefore, the solution set of $\mathbf{Ax} = \mathbf{0}$ contains non-trivial solutions. On the other hand, suppose instead that $p < n$. In Lecture 4, Theorem 4.11, we proved that a set of vectors $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ in $\mathbb{R}^n$ spans $\mathbb{R}^n$ if and only if the RREF of $\mathbf{A}$ has exactly $r = n$ leading ones. The largest possible value of r is $r = p < n$. Therefore, if $p < n$ then $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ cannot be a basis for $\mathbb{R}^n$. Thus, in either case ($p > n$ or $p < n$), the set $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ cannot be a basis for $\mathbb{R}^n$. Hence, any basis in $\mathbb{R}^n$ must contain n vectors. $\square$

The previous theorem does not say that every set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ of nonzero vectors in $\mathbb{R}^n$ containing n vectors is automatically a basis for $\mathbb{R}^n$. For example,

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$$

do not form a basis for $\mathbb{R}^3$ because

$$\mathbf{x} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

is not in the span of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. All that we can say is that a set of vectors in $\mathbb{R}^n$ containing fewer or more than n vectors is automatically **not** a basis for $\mathbb{R}^n$. From Theorem 16.10, any basis in $\mathbb{R}^n$ must have exactly n vectors. In fact, on a general abstract vector space V, if $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for V then any other basis for V must have exactly n vectors also. Because of this result, we can make the following definition.

Definition 16.11: Let V be a vector space. The **dimension** of V, denoted $\dim \mathsf{V}$, is the number of vectors in any basis of V. The dimension of the trivial vector space $\mathsf{V} = \{\mathbf{0}\}$ is defined to be zero.

There is one subtle issue we are sweeping under the rug: Does every vector space have a basis? The answer is yes but we will not prove this result here.

Moving on, suppose that we have a set $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ in $\mathbb{R}^n$ containing exactly n vectors. For $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ to be a basis of $\mathbb{R}^n$, the set $\mathcal{B}$ must be linearly independent and $\operatorname{span} \mathcal{B} = \mathbb{R}^n$. In fact, it can be shown that if $\mathcal{B}$ is linearly independent then the spanning condition $\operatorname{span} \mathcal{B} = \mathbb{R}^n$ is automatically satisfied, and vice-versa. For example, say the vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ in $\mathbb{R}^n$ are linearly independent, and put $\mathbf{A} = [\mathbf{v}_1\ \mathbf{v}_2\ \cdots\ \mathbf{v}_n]$. Then $\mathbf{A}^{-1}$ exists and therefore $\mathbf{Ax} = \mathbf{b}$ is always solvable. Hence, $\operatorname{Col}(\mathbf{A}) = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} = \mathbb{R}^n$. In summary, we have the following theorem.

Theorem 16.12: Let $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be vectors in $\mathbb{R}^n$. If $\mathcal{B}$ is linearly independent then $\mathcal{B}$ is a basis for $\mathbb{R}^n$. Or if $\operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} = \mathbb{R}^n$ then $\mathcal{B}$ is a basis for $\mathbb{R}^n$.

Example 16.13. Do the columns of the matrix $\mathbf{A}$ form a basis for $\mathbb{R}^4$?

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & 3 & -2 \\ 4 & 7 & 8 & -6 \\ 0 & 0 & 1 & 0 \\ -4 & -6 & -6 & 3 \end{bmatrix}$$

Solution. Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ denote the columns of $\mathbf{A}$. Since we have $n = 4$ vectors in $\mathbb{R}^n$, we need only check that they are linearly independent. Compute

$$\det \mathbf{A} = -2 \neq 0$$

Hence, $\operatorname{rank}(\mathbf{A}) = 4$ and thus the columns of $\mathbf{A}$ are linearly independent. Therefore, the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ form a basis for $\mathbb{R}^4$. □

A subspace W of a vector space V is a vector space in its own right, and therefore also has dimension. By definition, if $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is a linearly independent set in W and $\operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} = \mathsf{W}$, then $\mathcal{B}$ is a basis for W and in this case the dimension of W is k. Since an n-dimensional vector space V requires exactly n vectors in any basis, then if W is a strict subspace of V then

$$\dim \mathsf{W} < \dim \mathsf{V}.$$

As an example, in $\mathsf{V} = \mathbb{R}^3$ subspaces can be classified by dimension:

1. The zero dimensional subspace in $\mathbb{R}^3$ is $\mathsf{W} = \{\mathbf{0}\}$.
2. The one dimensional subspaces in $\mathbb{R}^3$ are lines through the origin. These are spanned by a single non-zero vector.
3. The two dimensional subspaces in $\mathbb{R}^3$ are planes through the origin. These are spanned by two linearly independent vectors.
4. The only three dimensional subspace in $\mathbb{R}^3$ is $\mathbb{R}^3$ itself. Any set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ in $\mathbb{R}^3$ that is linearly independent is a basis for $\mathbb{R}^3$.

Example 16.14. Find a basis for $\operatorname{Null}(\mathbf{A})$ and the $\dim \operatorname{Null}(\mathbf{A})$ if

$$\mathbf{A} = \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix}.$$

Solution. By definition, the $\operatorname{Null}(\mathbf{A})$ is the solution set of the homogeneous system $\mathbf{A}\mathbf{x} = \mathbf{0}$. Row reducing we obtain

$$\mathbf{A} \sim \begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 1 & 5/2 & 3/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The general solution to $\mathbf{Ax} = \mathbf{0}$ in parametric form is

$$\mathbf{x} = t \begin{bmatrix} -5 \\ -3/2 \\ 0 \\ 1 \end{bmatrix} + s \begin{bmatrix} -6 \\ -5/2 \\ 1 \\ 0 \end{bmatrix} = t\mathbf{v}_1 + s\mathbf{v}_2$$

By construction, the vectors

$$\mathbf{v}_1 = \begin{bmatrix} -5 \\ -3/2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -6 \\ -5/2 \\ 1 \\ 0 \end{bmatrix}$$

span the null space $(\mathbf{A})$ and they are linearly independent. Therefore, $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis for $\mathrm{Null}(\mathbf{A})$ and therefore $\dim \mathrm{Null}(\mathbf{A}) = 2$. In general, the dimension of the $\mathrm{Null}(\mathbf{A})$ is the number of free parameters in the solution set of the system $\mathbf{Ax} = \mathbf{0}$, that is,

$$\dim \mathrm{Null}(\mathbf{A}) = d = n - \mathrm{rank}(\mathbf{A})$$

□

Example 16.15. Find a basis for $\mathrm{Col}(\mathbf{A})$ and the $\dim \mathrm{Col}(\mathbf{A})$ if

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & -4 & 8 \\ 1 & 2 & 0 & 2 & 8 \\ 2 & 4 & -3 & 10 & 9 \\ 3 & 6 & 0 & 6 & 9 \end{bmatrix}.$$

Solution. By definition, the column space of $\mathbf{A}$ is the span of the columns of $\mathbf{A}$, which we denote by $\mathbf{A} = [\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3\ \mathbf{v}_4\ \mathbf{v}_5]$. Thus, to find a basis for $\mathrm{Col}(\mathbf{A})$, by trial and error we could determine the largest subset of the columns of $\mathbf{A}$ that are linearly independent. For example, first we determine if $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent. If yes, then add $\mathbf{v}_3$ and determine if $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent. If $\{\mathbf{v}_1, \mathbf{v}_2\}$ is not linearly independent then discard $\mathbf{v}_2$ and determine if $\{\mathbf{v}_1, \mathbf{v}_3\}$ is linearly independent. We continue this process until we have determined the largest subset of the columns of $\mathbf{A}$ that is linearly independent, and this will yield a basis for $\mathrm{Col}(\mathbf{A})$. Instead, we can use the fact that matrices that are row equivalent induce the same solution set for the associated homogeneous system. Hence, let $\mathbf{B}$ be the RREF of $\mathbf{A}$:

$$\mathbf{B} = \mathrm{rref}(\mathbf{A}) = \begin{bmatrix} 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

By inspection, the columns $\mathbf{b}_1, \mathbf{b}_3, \mathbf{b}_5$ of $\mathbf{B}$ are linearly independent. It is easy to see that $\mathbf{b}_2 = 2\mathbf{b}_1$ and $\mathbf{b}_4 = 2\mathbf{b}_1 - 2\mathbf{b}_3$. These same linear relations hold for the columns of $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & -4 & 8 \\ 1 & 2 & 0 & 2 & 8 \\ 2 & 4 & -3 & 10 & 9 \\ 3 & 6 & 0 & 6 & 9 \end{bmatrix}$$

By inspection, $\mathbf{v}_2 = 2\mathbf{v}_1$ and $\mathbf{v}_4 = 2\mathbf{v}_1 - 2\mathbf{v}_3$. Thus, because $\mathbf{b}_1, \mathbf{b}_3, \mathbf{b}_5$ are linearly independent columns of $\mathbf{B} = \texttt{rref}(\mathbf{A})$, then $\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5$ are linearly independent columns of $\mathbf{A}$. Therefore, we have

$$\text{Col}(\mathbf{A}) = \text{span}\{\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5\} = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 8 \\ 8 \\ 9 \\ 9 \end{bmatrix} \right\}$$

and consequently $\dim \text{Col}(\mathbf{A}) = 3$. **This procedure works in general:** To find a basis for the $\text{Col}(\mathbf{A})$, row reduce $\mathbf{A} \sim \mathbf{B}$ until you can determine which columns of $\mathbf{B}$ are linearly independent. The columns of $\mathbf{A}$ in the same position as the linearly independent columns of $\mathbf{B}$ form a basis for the $\text{Col}(\mathbf{A})$.

WARNING: Do not take the linearly independent columns of $\mathbf{B}$ as a basis for $\text{Col}(\mathbf{A})$. Always go back to the original matrix $\mathbf{A}$ to select the columns. □

After this lecture you should know the following:

- what it means for a set to be linearly independent/dependents
- what a basis is (a spanning set that is linearly independent)
- what is the meaning of the dimension of a vector space
- how to determine if a given set in $\mathbb{R}^n$ is linearly independent
- how to find a basis for the null space and column space of a matrix $\mathbf{A}$